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ORIGINAL RESEARCH PAPER

IQ-VIO: adaptive visual inertial odometry via interference quantization under dynamic environments

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Abstract
Vision-based localization is susceptible to interference from dynamic objects in the environment, resulting in decreased localization accuracy and even tracking loss. Hence, sensor fusion with IMUs or motor encoders has been widely adopted to improve positioning accuracy and robustness in dynamic environments. Commonly used loose coupling fusion localization methods cannot completely eliminate the error introduced by dynamic objects. In this paper, we propose a novel adaptive visual inertial odometry via interference quantization, namely IQ-VIO. To quantify the confidence of pose estimation through vision frames analysis, we firstly introduce the feature coverage and the dynamic scene interference index based on image information entropy. Then, based on the interference index, we further establish the IQ-VIO multi-sensor fusion model, which can adaptively adjust the measurement error covariance matrix of an extended Kalman filter to suppress and eliminate the impact of dynamic objects on localization. We verify IQ-VIO algorithm on KAIST Urban dataset and actual scenes. Results show that our method achieves favorable performance against other algorithms. Especially under challenging scenes such as low texture, the RPE of our algorithm decreases at least twenty percent. Our approach can effectively eliminate the impact of dynamic objects in the scenes and obtain higher positioning accuracy and robustness than conventional methods.

Keywords Localization · Visual inertial odometry · Dynamic simultaneous localization and mapping (SLAM) · Extended Kalman filter (EKF)

1 Introduction

Due to the advantages of low cost and informativeness, visual simultaneous localization and mapping (SLAM) has received high attention and research efforts over the last years. However, most of the current approaches overly rely on static world assumption [1]. Dynamic objects and their motion amplitude can have significant adverse effect on positioning accuracy and even lead to loss of tracking. To address this issue, it has become a common solution to integrate the visual sensor with the inertial unit and the motor encoder [2].

General fusion localization methods fall into two categories: tight coupling [3] and loose coupling [4]. Existing loose coupling model generally uses the kinematic model of the wheel odometer to predict the robot pose and update the measured value of the visual odometer. However, this kind of method usually ignores the interference of dynamic objects on the visual odometer measurements and can hardly obtain ideal final positioning results. While in the case of tight coupling, there are higher requirements for initialization scenarios, which means serious positioning accuracy losses will happen if it is initialized in the dynamic scenarios. When a single vision sensor is used to solve the positioning problem in dynamic scenes, there are still some challenges, such as poor accuracy and possible failure of the algorithm in some scenes.

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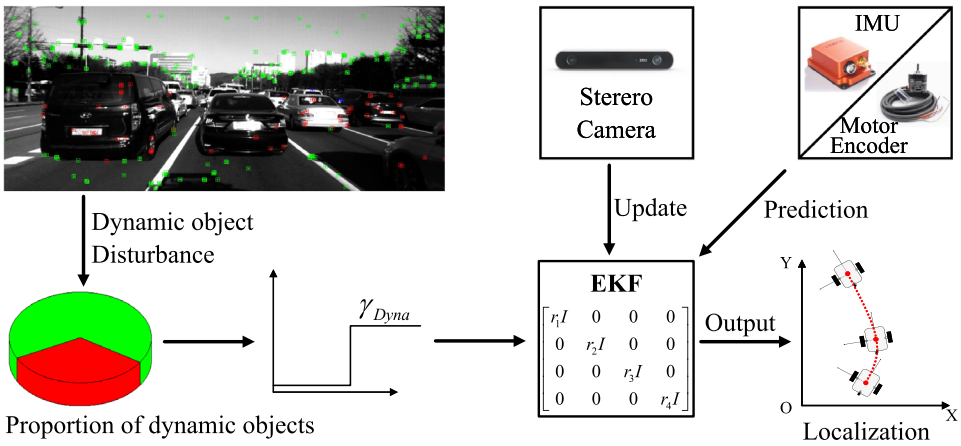


Fig. 1 The infographic of IQ-VIO

To address above problems, we propose a novel multi-sensor adaptive fusion method: adaptive visual inertial odometry via interference quantization (IQ-VIO). Figure 1 briefly illustrates the principle of this algorithm. Different from traditional Kalman filter fusion methods, our method detects and quantifies the interference of dynamic objects with the coverage of visual features, which is positively correlated with information entropy. Instead of masking all the dynamic objects like DynaSLAM [5], features upon dynamic objects are still utilized. We introduce a dynamic scene binary interference index, whose value changes according to the interference effects of dynamic objects on visual posture. In addition, we also determine a quantitative relationship between the interference index and the covariance matrix of visual pose measurement error in extended Kalman filter (EKF) through experimental heuristics. By dynamically affecting the measurement update process of EKF, the interference index contributes to better positioning accuracy in dynamic scenes and has higher robustness and accuracy than conventional methods. To summarize, our main contributions are as follows:

- Based on image information entropy, we introduce the coverage of visual features and a dynamic scene interference index to quantify the impact of dynamic objects in the scene on visual localization.
- Based on the interference index of dynamic scene, we propose an adaptive VIO algorithm under dynamic environments. Combining the interference index with EKF can effectively eliminate the impact of dynamic objects in scenes on the localization results.

This paper is organized as follows: Section 2 introduces the related works. Section 3 describes the establishment of the dynamic scene interference index. Section 4 describes the implementation of adaptive visual inertial odometry fusion algorithm. In Sect. 5, several experiments are designed to test the robustness and accuracy of IQ-VIO. Finally, some conclusions are drawn in Sect. 6.

2 Related work

Pure visual SLAM system in dynamic scenes can be divided into two categories: geometry-based [6] and deep learning-based [7].

Geometry-based motion segmentation algorithms generally determine the presence of dynamic objects in the scene by multi-view geometry [8], reprojection error [9], optical flow [10], etc. This kind of method has good real-time performance, but it cannot handle degenerate motion. When the dynamic objects and the visual sensor remain relatively stationary, the dynamic objects are easy to be taken as part of the static scene. In addition, when the proportion of dynamic areas in the image reaches more than half, the geometry-based methods cannot distinguish between static and dynamic parts of the image. Li et al. [11] use the static weight of the object edge points to represent the possibility of stationary, and combine the weight to segment the edge points of the moving object in intensity-assisted iterative closure point (IAICP) to reduce the impact of the moving object on pose estimation. The algorithm can run in real time but requires a large number of depth edges in the environment. Atashgah et al. [12, 13] obtain real-time images with a monocular camera and further combine them with the trajectory predicted by aircraft dynamics in the deblurring-in-the-loop single camera Simultaneous Localization and Mapping (Mono-SLAM). Derome [14] uses the residual between the predicted image and the observed image to calculate the optical flow, and identifies the dynamic object by detecting the residual field. Wang et al. [15], aiming at the poor performance of RGB-D SLAM [16] in dynamic scenes, introduce optical flow extraction and sparse point trajectory clustering to segment dynamic objects. Alcantarilla et al. [17] judge the moving object through the calculation result of the

likelihood of the residual motion. If the residual is very low, the corresponding features are likely to be located on the static object.

With the development of deep learning, the accuracy of target recognition and semantic segmentation has been greatly improved. Yu et al. [18] add a semantic segmentation module to ORBSLAM2 [1] and propose DS-SLAM. The algorithm combines semantic segmentation network with mobile consistency check method to filter the dynamic features, and finally tracks the remaining static features. However, the algorithm performs poorly in low dynamic sequences, and the object of semantic segmentation is limited to human. Based on ORBSLAM2 and semantic segmentation, researchers propose new algorithms such as Detect-SLAM [19] and Dynamic-SLAM [20] and achieve good results. The advantage of the method combined with semantic segmentation is that it can accurately recognize or segment dynamic objects through high-dimensional information in the image [21, 22]. Combined with traditional SLAM, this kind of method can significantly improve the positioning accuracy, but its disadvantages are also obvious, i.e., high requirements for hardware, vulnerable to the influence of image blur, and unable to recognize untrained dynamic. In addition, due to the large proportion of dynamic objects in some scenes, small part of static area is left in the images for visual estimation after segmentation, leading to the failure of the algorithm. In recent years, researchers demonstrate the feasibility of estimating pose directly using deep learning network, i.e., GeoNet [23] and DeepVO [24]. However, this kind of method is in lack of interpretability and needs excessive pre-training time. In addition, this kind of algorithm also suffers from generalization problems.

Combining multiple sensors can effectively improve the performance of the SLAM system. VI-SLAM is usually composed of visual sensors and IMUs. For example, RTAB-MAP [25] is a versatile system, which can robustly combine RGB-D images with wheel odometry. However, it cannot handle dynamic environments. VINS [26] has certain robustness to dynamic objects, but it has high requirements for initialized scenes. This paper focuses on the visual inertial odometry to estimate the trajectory of the device. Babu et al. [27] propose a method to detect the conflict between vision-only and inertial-only estimation, but it assumes that inertial measurement is always more reliable. VIODE [28] uses semantic segmentation and removes dynamic objects to better estimate ego-motion, but this approach requires enumeration of static and dynamic classes. Wu et al. [29] propose a preprocessing VIO step for reducing influence of lines on dynamic objects, and merge it into a VIO pipeline. GMSK-SLAM [30] adopts Grid-based Motion Statistics feature points matching method and K-means cluster algorithm to distinguish the dynamic objects. YO-VIO [31] removes the feature points on dynamic objects combining semantic object detection and odometry, but using odometry in the front-end influences real-time performance.

Table 1 The frequently used symbols throughout the article

Meaning	Symbol
Dynamic scene interference index	γ_{Dyna}
Feature coverage	C_{f}
Depth threshold	β
C_{f} of features within the depth range $(0, \beta)$	C_{fn}
C_{fn} of the dynamic objects	C_{fd}

3 Confidence analysis of visual pose and establishment of the interference index

Focusing on the visual front-end of the feature-based method, this section introduces a method to quantify the amount of information of each image frame in the visual tracking process, so as to establish the dynamic scene interference index. The frequently used symbols throughout the article are listed in Table 1 for ease of reading.

3.1 Feature coverage

For the feature-based front-end visual localization algorithm, the accuracy of the algorithm depends on both of the quality of feature extraction and the accuracy of feature matching. Features are another kind of mathematical representation of image information. For each frame of image, the number and the uniformity of extracted features will affect the accuracy of pose tracking. For example, ORBSLAM2 uses a descriptor with anti-noise ability and rotation invariance to quickly extract and describe features, so as to reduce the aggregation of features, and further improve the tracking accuracy. Many scholars [32] study the extraction and distribution. They have directly or indirectly proved the proportional relationship between the coverage of features and the amount of information contained in the image. The information contained in the image can be called information entropy $H_\alpha(f)$. The greater the information entropy in an image, the more information it contains. Therefore, we define a metric called feature coverage, which is proportional to the information entropy, to evaluate the information contained in the features of each image frame. We need to obtain the minimal connected subgraph length first:

All features in an image frame form a point set $V = \{P_i\}_{i=1}^{n}$, where $P_i = (u_i, v_i)$ is the coordinates of the matched points in the image frame. Points in the point set are connected to form a non-connected graph, and all edges in the graph form an edge set $E = e_{ij} = (P_i, P_j)|i, j = 1, 2, ...,$

$n, i \neq j$, where e_{ij} is the distance between P_i and P_j. Our goal is to obtain a final minimal connected subgraph length

$$L = \min \sum_{e} |e|, \tag{1}$$

where e is the edge of the minimum spanning tree (MST) which consists of the feature points of the target area. For the target area in the image, we firstly establish an empty point set V_{new} and V_{goal}, where V_{goal} is the point set of near points in this area. Then, we extract P_i in V_{goal} and search for the minimal edge $e_{ij} = (P_i, P_j)$, where P_j also belongs to V_{goal}. The length of the minimal connected subgraph is initialized to $L = e_{ij}$, and then, we add P_i and P_j to point set V_{new}. The edge $e_{nm} = (P_n, P_m)$ with the smallest weight is continuously selected from the edge set E, where $P_n \in V_{\text{new}}$, $P_m \notin V_{\text{new}}$, which is used to update the value of L making $L = L + e_{nm}$. Then, P_m is added to the point set V_{new}, until V_{new}=V_{goal}, and the final minimal connected subgraph length L is obtained.

For the image with length l and height h, when the number of features extracted and matched is n, the feature coverage can be defined as:

$$C_{\text{f}} = \frac{\lambda L}{n^{\alpha}(l + h - 2E(X))}, \tag{2}$$

where L is in direct proportion to the information entropy γ_{Dyna} [33], and the proportional relationship is:

$$H_{\alpha}(f) \propto \frac{1}{1-\alpha} \log\left(\frac{L}{n^{\alpha}}\right). \tag{3}$$

Therefore, the relationship between $H_{\alpha}(f)$ and C_{f} is: $H_{\alpha}(f) \propto C_{\text{f}}$. α and λ are weighting coefficients. Generally, α is 0.5 [34] and λ is a heuristic normalized parameter set at 1.4. For the whole image, the feature coverage, thus, can be a metric for global information.

The image frame size is different in actual visual tracking process. In order to make the work more versatile, the expected value $E(X)$ of the distance between random points in the image is introduced as a division factor in (2). Assuming that the length and width of the image are l and h, the expected value of the distance between any two points $A(x, y)$ and $B(u, v)$ in the image can be expressed as:

$$E(X) = \int_0^l \int_0^h \int_0^l \int_0^h \sqrt{(x-u)^2 + (y-v)^2}\,\mathrm{d}x\mathrm{d}y\mathrm{d}u\mathrm{d}v. \tag{4}$$

In order to verify that (2) can reflect the coverage of visual feature matching points to a certain extent, the randomly generated two-dimensional point set is used to simulate the extraction of features in the actual visual tracking process. Figure 2 shows three randomly generated two-dimensional point sets within the range of $x \in [-1, 1]$ and $y \in [-1, 1]$, where the number of randomly generated points is 200.

We calculate the feature coverage of the point sets of these three distributions and illustrate in Fig. 3, where we show that the feature coverage C_{f} can directly reflect the coverage of randomly generated points to a certain extent, which will not fluctuate with the number of generated points and the aspect ratio of the generated area. The experiment results are very stable within a regular number of visual tracking feature matching points. The calculation time is between 1ms and 3ms and increases linearly with the number of generated points.

3.2 Interference index

To use an adaptive algorithm to eliminate the interference of dynamic objects in the scene, the dynamic objects must first be identified and segmented. Dynamic objects detection is based on deep learning which utilizes semantic information to identify and calculate the dynamic features in the scene. The target detection network used is YOLOv4 [35], and the training set uses the MS coco dataset, with a total of 80 kinds of targets. Considering our experiment scenes, the algorithm focuses on segmenting dynamic objects, pedestrians, vehicles and bicycles.

Given an image as input, YOLOv4 identifies the dynamic objects and segments the objects range with bounding boxes, and finally the feature points above are exploited to calculate the dynamic scene interference index. YOLOv4 weight parameters use AlexeyAB open-source pretrained weight values (https://github.com/AlexeyAB/darknet).

The framework of visual tracking in this paper is roughly the same as ORBSLAM2. On the basis of ORBSLAM2, a semantic segmentation module of dynamic objects is added. The framework is shown in Fig. 4.

After the camera has collected multiple frames of images, the semantic segmentation module will extract the dynamic objects in the image frame and create a new mask, while the areas in the original image with dynamic objects will be marked. At the same time, the visual tracking module will extract the features of the image frame, calculate the disparity between the left and right eyes, and compute the depth information of each matching feature through the disparity.

Then, we calculate the coverage C_{f} of the feature matching points within the depth range $(0, \beta)$ to obtain reliable estimation. For binocular ORBSLAM2 algorithm, the stereo images are normalized and matched by sum of absolute differences (SAD). The smaller the SAD value, the lower the error rate of the point matching. When the SAD value is greater than a certain threshold, the point is regarded as an outlier, and the depth of the point is set to -1. Therefore, if the depth information of a feature matching point is less

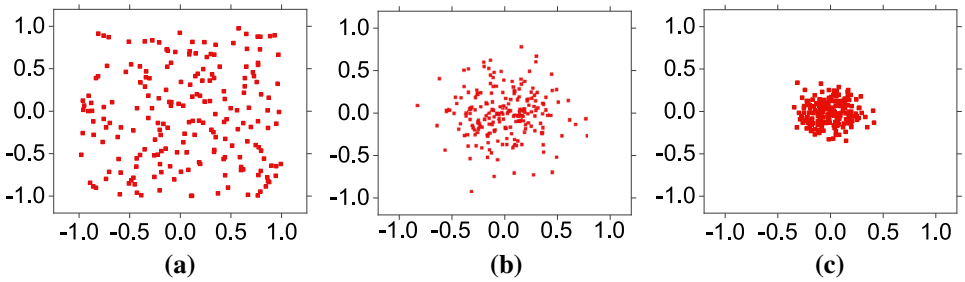


Fig. 2 Two-dimensional random point sets with different distributions. **a** Uniform distribution, **b** Gaussian distribution (variance of 0.3), **c** Gaussian distribution (variance of 0.1)

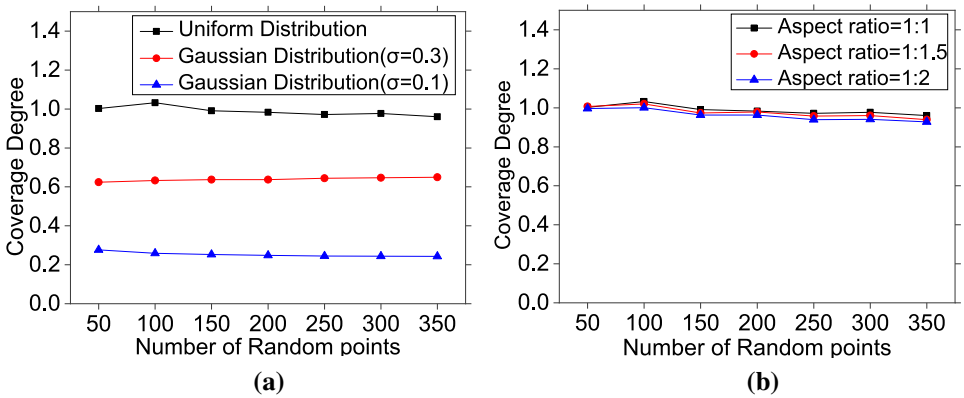


Fig. 3 Coverage curves of random point sets with different distributions and aspect ratios. **a** Different distribution, **b** Different aspect ratios of uniform distribution

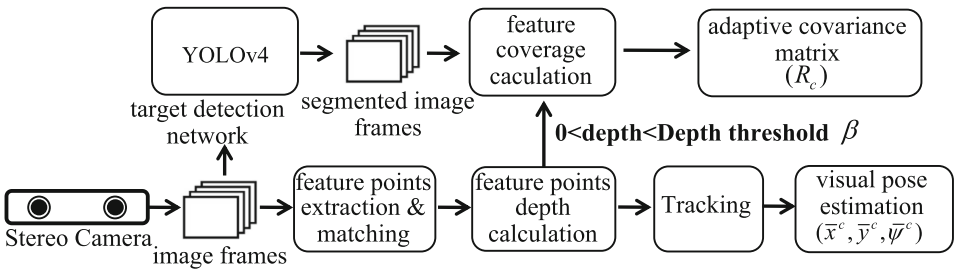


Fig. 4 Procedure of visual pose estimation

than zero, it indicates that the error of the feature in matching is large, and such points are generally excluded in the subsequent calculation. When the depth information is greater than the depth threshold β, it indicates that the parallax of the feature matching point on the stereo image is too small to reliably estimate its depth. Generally, such features will only have a partial influence on pose estimation. The feature matching points whose depth is within the depth range $(0, \beta)$ are called near points, and matching features whose depth is greater than the depth threshold are called as far points.

Therefore, C_{fn} is obtained by calculating the C_{f} of the feature set composed of near points in the whole image, and C_{fd} is obtained by calculating the C_{f} of feature set composed of near points on dynamic objects. Based on these two indicators, we define a dynamic scene interference index γ_{Dyna}, whose corresponding relationship is:

$$\gamma_{\text{Dyna}} = \begin{cases} 1, & \frac{C_{\text{fd}}}{C_{\text{fn}}} > 0.3 \\ 1, & C_{\text{fn}} < 0.28 \\ 0, & \text{else} \end{cases} \tag{5}$$

Among them, 0.3 and 0.28 are heuristic parameter values continuously adjusted after the algorithm runs in each sequence. $\frac{C_{\text{fd}}}{C_{\text{fn}}}$ represents the proportion of dynamic objects in the scene, and if it is high, it will cause the visual odometry to be inaccurate. If C_{fn} is too small, it means that there are a small amount of near points in the environment where the camera is located, so the estimated pose is more likely to contain errors.

The visual tracking thread continues to run and finally obtain the estimated value of visual pose, and meanwhile, the measurement error covariance matrix of visual pose is adaptively adjusted according to the calculated dynamic scene interference index γ_{Dyna}, which is used in the EKF fusion in Sect. 4.

4 System model

The overall fusion positioning architecture proposed is shown in Fig. 5. Given that a mobile robot moves in a two-

dimensional plane, the displacement in the Z direction, pitch angle, roll angle, and the speed and angular speed in the corresponding direction are all expressed as 0.

First, we define the twelve-dimensional state variables x and the observation z:

$$x = \left[\vec{p}\ \vec{\alpha}\ \vec{v}\ \vec{w} \right]^T, \tag{6}$$

$$z = \left[\vec{\tilde{p}}\ \vec{\tilde{\alpha}}\ \vec{\tilde{v}}\ \vec{\tilde{w}} \right]^T. \tag{7}$$

where $\vec{p}$ is the three-axis displacement, $\vec{\alpha}$ is the three-axis attitude angle, $\vec{v}$ is the three-axis velocity, and $\vec{w}$ is the three-axis angular velocity. Then, we initialize the average value of the state variables and the error covariance matrix with the data at time t_0:

$$\hat{x}_0 = E(x_0), \tag{8}$$

$$P_0 = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T]. \tag{9}$$

Without considering the influence of external control factors on the robot, the state variables are predicted and updated as:

$$\begin{aligned} \hat{x}_{t|t-1} &= f(\hat{x}_{t-1|t-1}) \\ &= \left[\vec{p}_{t-1} + \vec{v}_{t-1}\Delta t\ \vec{\alpha}_{t-1} + \vec{w}_{t-1}\Delta t\ \vec{v}_{t-1} + \vec{a}_{t-1}\Delta t\ \vec{w}_{t-1} \right]^T. \end{aligned} \tag{10}$$

It is not a linear equation, in which the displacement is predicted and updated by the posterior velocity of the last time. Since the displacement data in the state variables is in the world coordinate system, the variables in the following formulas are marked with their coordinate system in the upper right corner. w represents the world coordinate system, and b represents the body coordinate system, so (11) can be obtained through coordinate transformation. The velocity at this time is predicted and updated by the acceleration in the inertial measurement unit as (12). The yaw is predicted by the angular speed of body calculated by the motor encoder as (13):

$$\begin{cases} p^w_{x,t} = p^w_{x,t-1} + v^b_{x,t-1}\cos\alpha^w_{z,t-1}\Delta t \\ p^w_{y,t} = p^w_{y,t-1} + v^b_{x,t-1}\sin\alpha^w_{z,t-1}\Delta t \end{cases} \tag{11}$$

$$v^b_{x,t} = v^b_{x,t-1} + a^b_{x,t-1}\Delta t, \tag{12}$$

$$\alpha^w_{z,t} = \alpha^w_{z,t-1} + w^b_{z,t-1}\Delta t. \tag{13}$$

The state transition matrix in the form of Jacobi matrix can be obtained from the partial derivative of the state transition equation:

$$F_t = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -v^b_{x,t-1}\sin\alpha^w_{z,t-1}\Delta t & \cos\alpha^w_{z,t-1}\Delta t & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & v^b_{x,t-1}\cos\alpha^w_{z,t-1}\Delta t & \sin\alpha^w_{z,t-1}\Delta t & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{14}$$

For posteriori measurement updates, the displacements in the twelve-dimensional measured variables are estimated by the visual tracking algorithm. The three-dimensional attitude angle will be measured and updated twice, once by the AHRS model and once by the visual tracking algorithm. The linear speed and angular speed are calculated and measured by the motor encoder of the chassis. Through the optimal estimation of the state variables at time t, since the measured variables and the state variables are both twelve dimensional vectors, the observation model matrix is a square matrix as (15) and the covariance matrix of observation noise is as (16):

$$H_t = \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix}, \tag{15}$$

$$R_t = \begin{bmatrix} r_1 I & 0 & 0 & 0 \\ 0 & r_2 I & 0 & 0 \\ 0 & 0 & r_3 I & 0 \\ 0 & 0 & 0 & r_4 I \end{bmatrix}, \tag{16}$$

where r_2–r_4 are the constant values obtained after the parameter adjustment, and r_1 is the dynamic parameter, whose size changes with the change of the dynamic scene interference index γ_{Dyna} proposed in the previous section:

$$r_1 = \begin{cases} e^{-6}, & \gamma_{\text{Dyna}} = 0, \\ e^{8}, & \gamma_{\text{Dyna}} = 1. \end{cases} \tag{17}$$

5 Experiments

All experiments are carried out on the ROS platform. The localization effect of IQ-VIO is compared with that of algorithms, including the traditional feature-based visual SLAM method—ORBSLAM2, the dynamic visual SLAM method based on deep learning—DynaSLAM, and the traditional

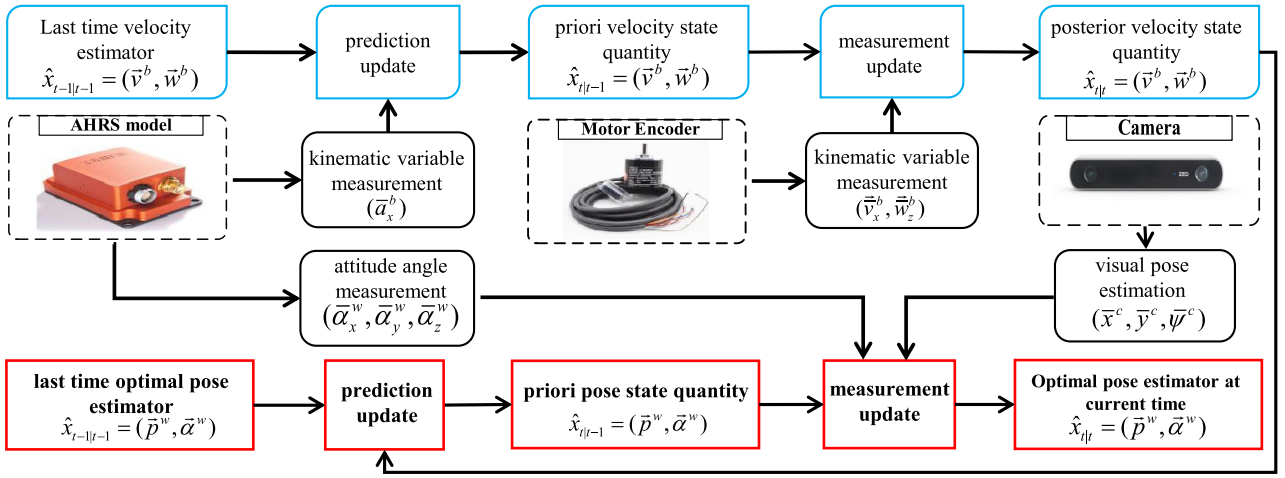


Fig. 5 Fusion positioning architecture of IQ-VIO

EKF fusion algorithm that combines ORBSLAM2. The parameters of ORBSLAM2 module are original fine-tuned and consistent across all algorithms, including the number of features, scale levels, and scale factor.

5.1 Experimental platform

The experimental platform in this paper is a differential mobile robot, as shown in Fig. 6. Our mobile robot is equipped with a 3D LIDAR sensor (velodyne VLP 16) and a stereo camera (stereolabs ZED) for perceiving information from environment, as well as navigation sensors such as AHRS module (xsense MTI 710) and RTK measuring instrument (Sinan Navigation M900SE-W), and master controller NVIDIA Xavier platform (Ubuntu 18.04 (aarch)). The EtherCAT slave station is connected with the motor encoder and responsible for transmitting the wheel odometer data to the Xavier platform as the master station. We primarily rely on three sensors for SLAM: a stereo camera, an AHRS module and a wheel odometer. The 3D LIDAR sensor and RTK serve as ground truth for the indoor and outdoor experiments.

In the indoor scene, LIO-SAM [36] algorithm will be used as the pose reference, which receives the data of 3D lidar and AHRS module. In the outdoor scene, RTK module M900SE-W is used to calculate the absolute pose reference. The positioning accuracy of these two pose reference systems has reached 1 cm.

5.2 Stationary experiment

In this experiment, the robot is placed in an office scene, and the robot remains stationary, which is similar to the dynamic sequence of the TUM dataset. In the scene, as shown in Fig. 7, there are two people sitting on the chair, talking and moving slowly, while the camera on the robot will record

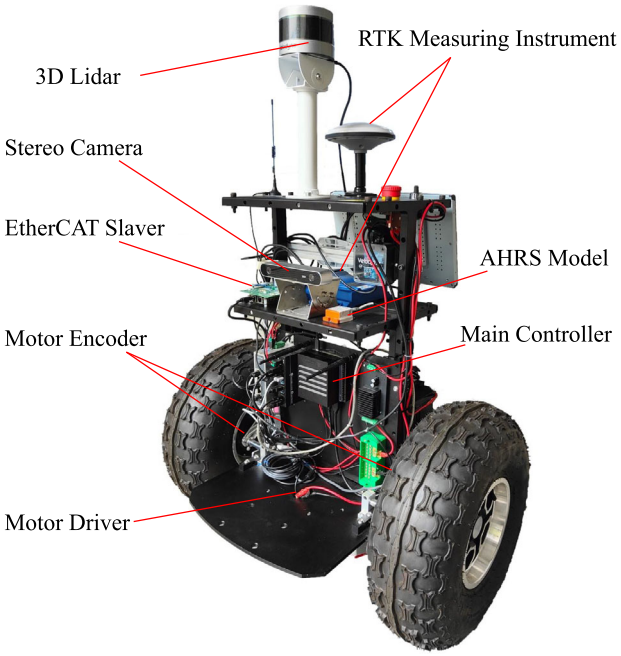


Fig. 6 Experimental platform

the scene. The purpose of this experiment is to evaluate the identification effect of our localization algorithm on dynamic objects and its robustness under the interference of dynamic objects. Because the wheel odometer is absolutely accurate, the visual pose is adaptively fused according to the confidence calculated by the dynamic object, which makes the final positioning result of this algorithm reach an accuracy level.

The visual pose confidence during the experiment is shown in Fig. 8. In the whole process of visual tracking, the ratio of dynamic objects coverages to the near-point coverage $\frac{C_{\text{fd}}}{C_{\text{fn}}}$ is greater than 0.3, which indicates that the whole process



Fig. 7 Stationary experiment scene

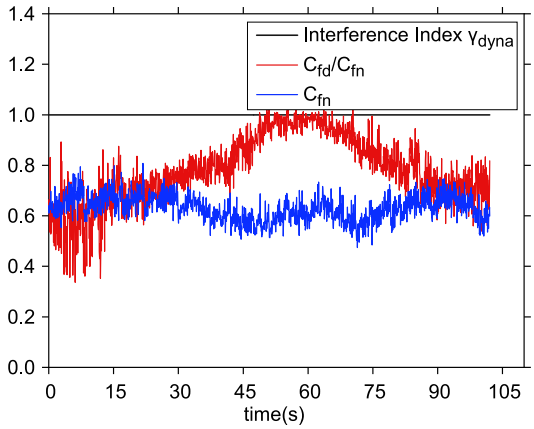


Fig. 8 Interference index and feature coverage in the stationary experiment

is affected by the dynamic object. Therefore, the corresponding interference index γ_{Dyna} is always 1, meaning that the confidence of visual pose decreases.

The trajectory of each algorithm in this scenario is shown in Fig. 9. Since the robot is always in a static state, the reference trajectory is the origin, and ORBSLAM2 has significant drift. As for the traditional EKF algorithm which fuses ORBSLAM2 and other sensors data, the value of visual pose is still be used to update the state in the filtering process, resulting in a certain degree of drift. Although DynaSLAM directly removes the pixels of dynamic objects, it fails to fully identify the dynamic object for a short period of time, which eventually leads to a little drift. For our algorithm, the confidence of the visual pose is always at a low level, so only the predicted speed of the wheel odometer is trusted during the state update process, and no drift occurs. The specific RMSE values of the four algorithms are shown in Table 2.

5.3 Evaluation on KAIST urban dataset

KAIST Urban dataset [37] records data in several cities in South Korea with complex environments, e.g., large complexes, metropolitan areas and underground car parks, which contains multiple sensor data such as LiDAR, GPS, vision and IMU data. It also includes ground truth generated by SLAM. Our goal is to achieve robot localization under the

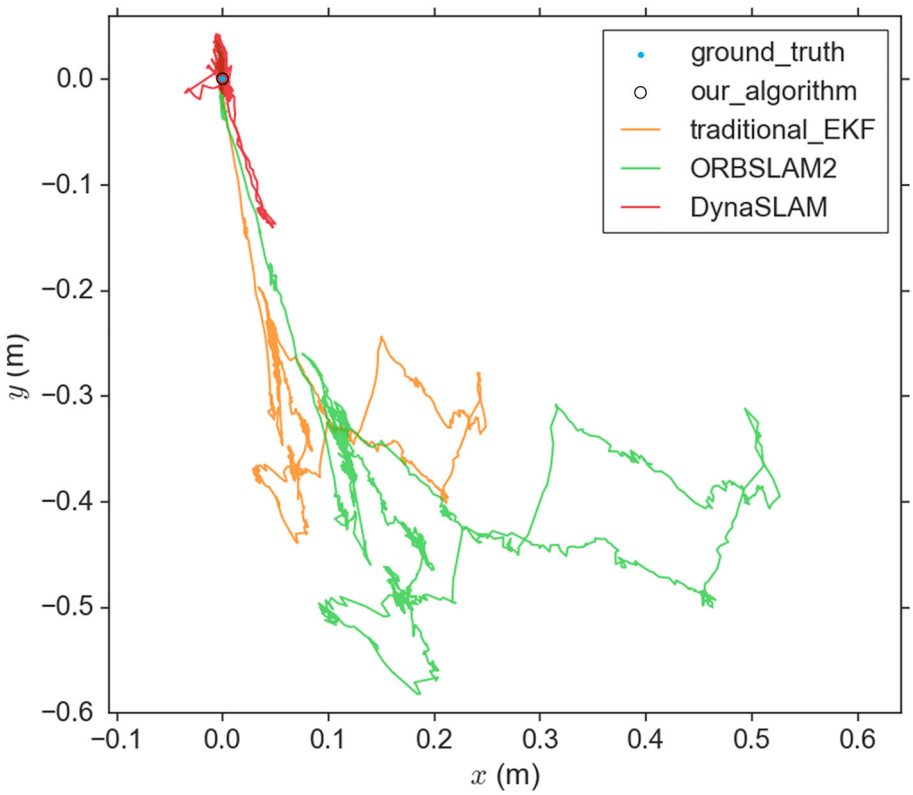


Fig. 9 The trajectories of different algorithms in the stationary experiment. The robot stays at the origin (the ground truth), and the robot trajectory of our algorithm is very close to the origin

Table 2 Accuracy evaluation of algorithms in the stationary experiment

Evaluate method (RMSE)	IQ-VIO (ours)	Traditional EKF	ORBSLAM2	DynaSLAM
Displacement RPE (m)	**1.12E−16**	0.00283	0.00673	0.00499
Angle RPE (°)	**0.00012**	0.00504	0.0942	0.0715
Displacement APE (m)	**1.18E−10**	0.301	0.43	0.0281

Best results are in bold

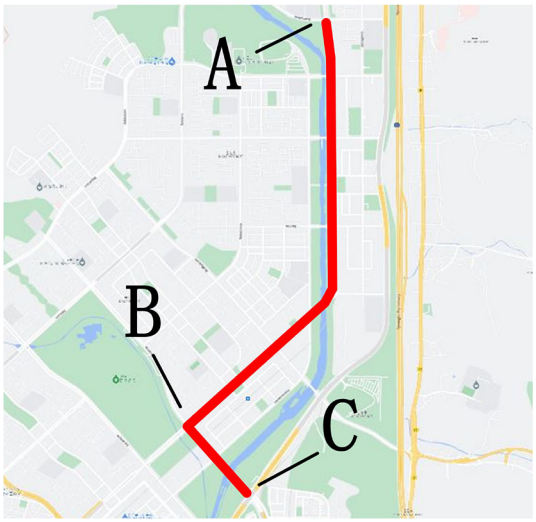


Fig. 10 Reference trajectory of Sequence 29. Point A is the start point of the dataset, point C is the end point of the dataset, and point B is a traffic intersection with dense vehicles. Segment A–B are low dynamic scenes and segment B–C are high dynamic scenes

impact of dynamic objects by multi-sensor fusion on KAIST Urban dataset.

We evaluate IQ-VIO in multiple sequences of the dataset. Since the dataset is located in the urban road, the vehicles and the pedestrians are potential dynamic objects, which will have a great impact on the visual algorithm. For example, when the speed of vehicles ahead is the same as ours during driving, the normal visual algorithm cannot exclude the dynamic features in the image frame through multi-view geometry, leading to wrong pose estimation. The sequence 29 is recorded in the urban area of Pankyo, South Korea, as shown in Fig. 10. The performance of the algorithms in low dynamic and high dynamic scenes can be analyzed simultaneously through this sequence. During the experiment, we found that when the vehicle is at the intersection of point B, those dense potential dynamic objects have a particularly severe impact on the visual estimation algorithm.

The trajectory obtained by each algorithm on this sequence is shown in Fig. 11. Our algorithm is the closest to the ground truth, and the trajectories of the two vision-based algorithms have different degrees of offset.

The circular magnifying boxes show the trajectories of the four algorithms at traffic intersection B, respectively. The positioning results of ORBSLAM2 have a huge drift, while DynaSLAM has no obvious drift on the trajectory because it directly removes the features on the dynamic objects. Despite avoiding drift, DynaSLAM estimates the vehicle position and pose while features are on distant objects, leading to large deviation. For our algorithm, since the dynamic measurement error covariance matrix is used in the filtering process, the confidence of the measured value of visual pose at point B is greatly reduced, and therefore, the impact of dynamic objects on pose estimation is successfully eliminated. Although the traditional EKF algorithm reduces the impact of dynamic objects to a certain extent due to the introduction of the IMU and other sensors, it cannot completely eliminate the errors.

The first image in Fig. 12a illustrates the features matching of DynaSLAM at point B. After removing the dynamic objects, almost 1/2 of the whole image is covered by the mask. Therefore, the feature extraction and matching focus on the distant traffic lights, street trees, and even the sky. When the intersection is resumed and the vehicle turns, it is very easy to cause tracking failure. In the cases of no tracking failure, these low-quality features have a great impact on the estimation of visual attitude, leading to a great deviation.

The first image in Fig. 12b presents the features of ORBSLAM2. Because a large number of features are concentrated on vehicles waiting for traffic signals, the vehicles in front and their own vehicle will remain relatively stationary when the intersection resumes traffic. As a result, the visual algorithm cannot correctly estimate the movement of its own position, which causes a large deviation in position estimation.

The first image in Fig. 12c shows the matching of features of IQ-VIO. The dynamic features whose depth value is less than the dynamic threshold and greater than zero are represented in red, and other features are represented in green. Only the red features will be used to calculate the dynamic covariance.

Figure 13 exhibits the coverage of the near points C_{fn} and the calculated proportion of dynamic objects $\frac{C_{\text{fd}}}{C_{\text{fn}}}$ in high dynamic scenes. The proportion of dynamic objects can directly reflect the influence of dynamic objects in the image frames during the visual tracking.

Figure 14 shows the position accuracy of each algorithm on the sequence. As shown in Fig. 14a, the performance of the four algorithms in low dynamic scenarios is quite similar. However, both DynaSLAM and ORBSLAM2 in Fig. 14b have large errors at 380 s (i.e., when the traffic intersection is resumed). The trajectory of DynaSLAM in Fig. 11 does

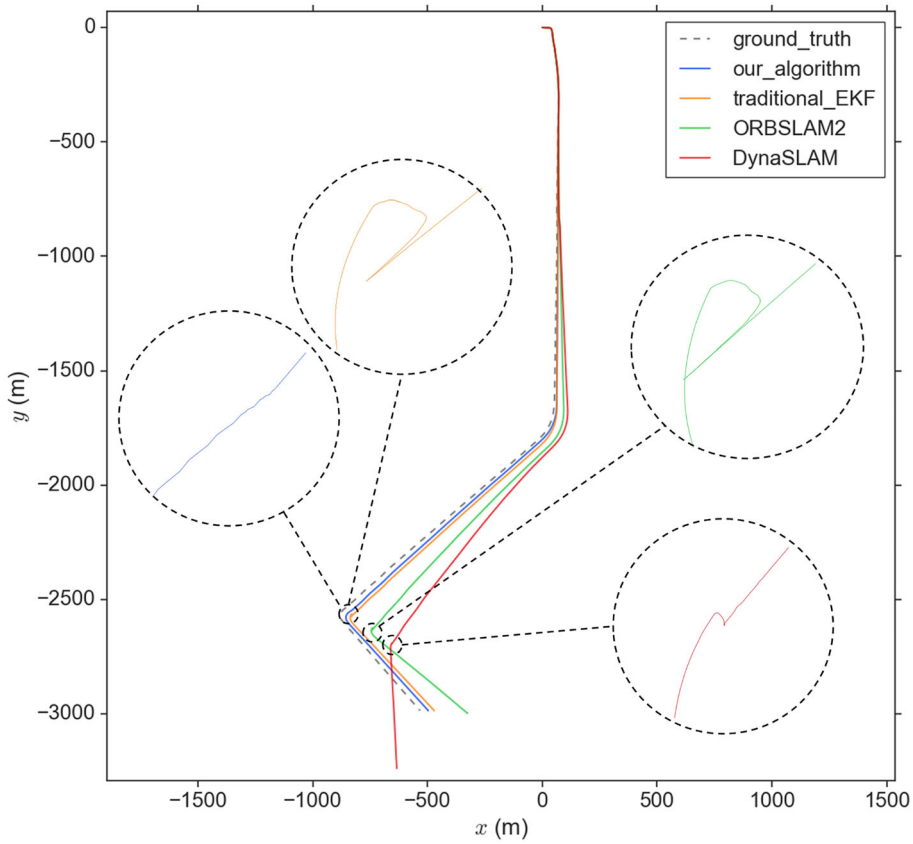


Fig. 11 Trajectories under sequence 29

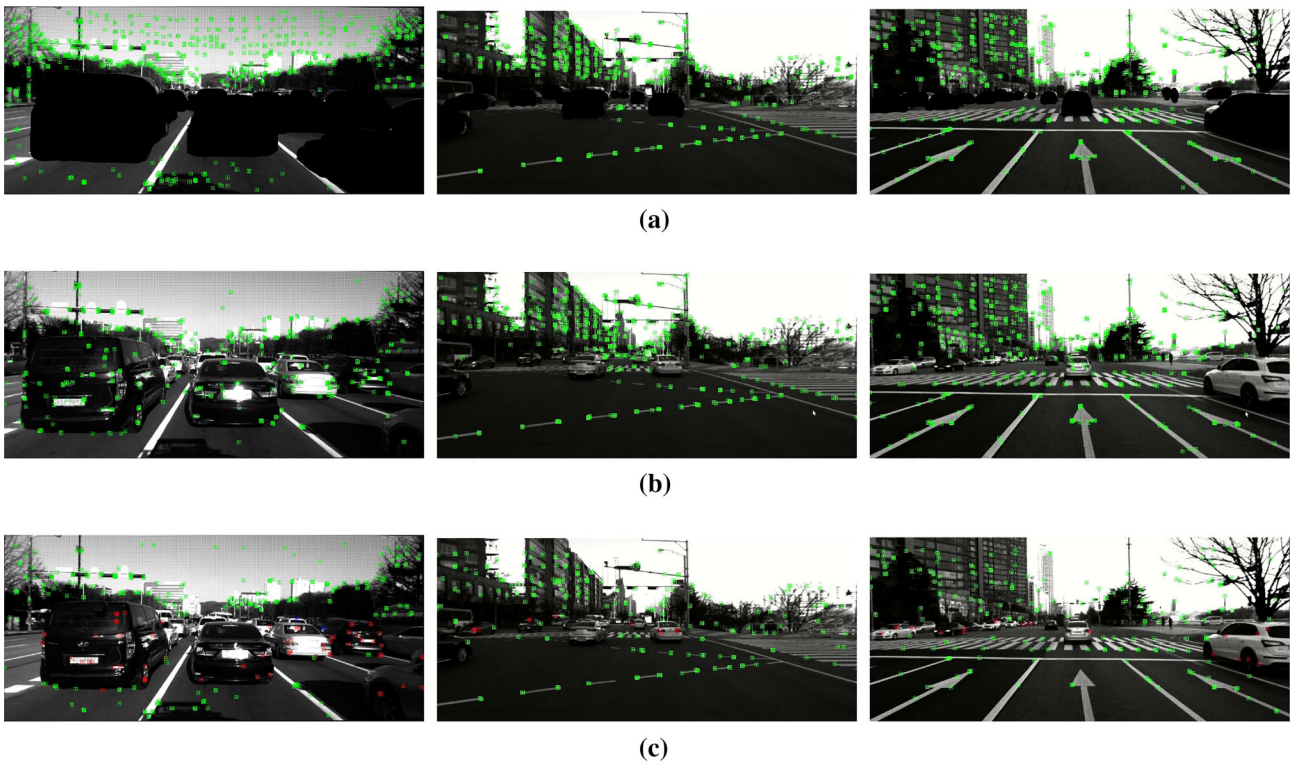


Fig. 12 The features of three algorithms in sequence 29. **a** DynaSLAM, **b** ORBSLAM2, where all the features are taken into account. **c** IQ-VIO, where the dynamic features within the depth range $(0, \beta)$ are represented in red, and others are represented in green (colour figure online)

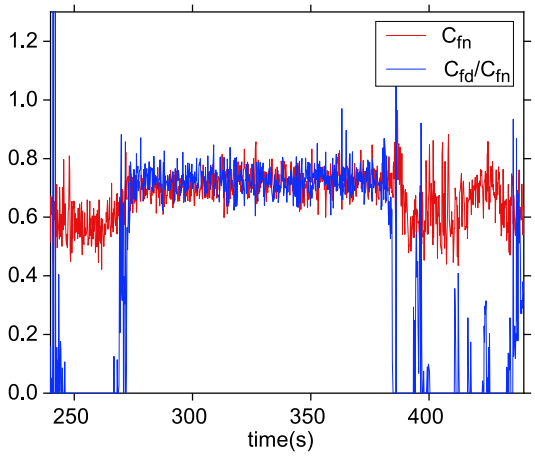


Fig. 13 Coverage and the calculated proportion of dynamic objects in high dynamic scenes of the near points

not show significant drift, because the position and pose estimation result of the vehicle remain stationary while features are on distant objects. However, the position of the vehicle is actually moving forward, so it still shows large deviation in Fig. 14b. In addition, during the experiments, DynaSLAM once fails to track because of poor quality of features upon distant object. Although the traditional EKF algorithm reduces the error at point B, the error elimination is not complete. The algorithm in this paper gets the optimal result here.

Extensive experiments are carried out under sequence 26/29/30/33/35 of KAIST urban dataset. Experimental results are shown in Table 3. Among them, the vehicle traffic flow in sequence 26 is not particularly dense, so each algorithm has achieved similar results. DynaSLAM also eliminated some

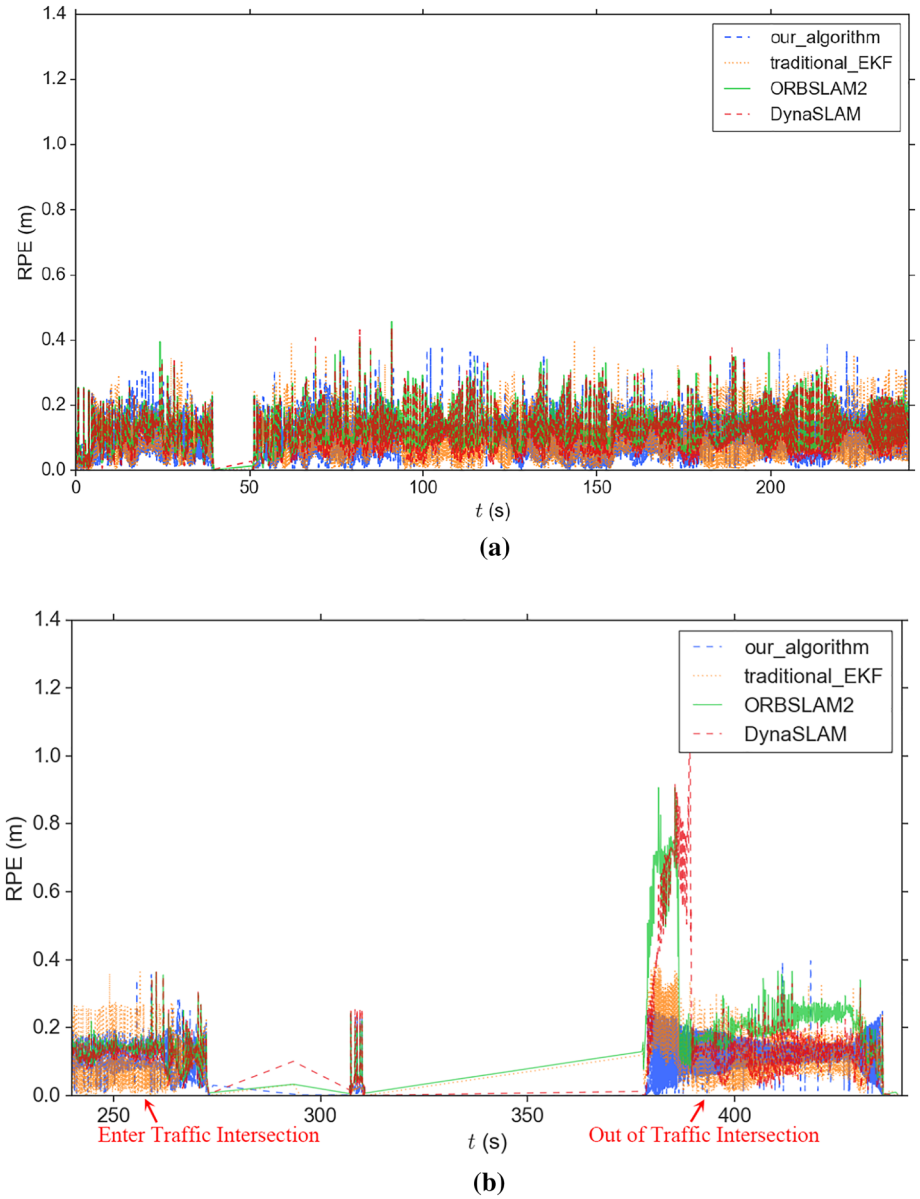


Fig. 14 Displacement RPE under sequence 29. **a** The low dynamic part of segment A–B within 0–240 s, **b** the high dynamic part of segment B–C within 240–440 s

Table 3 RPE and APE on KAIST urban dataset

Sequence number	IQ-VIO (ours)				Traditional EKF				ORBSLAM2				DynaSLAM			
	RPE (10^{-2} m)	APE (m)	RPE (10^{-2} deg)		RPE (10^{-2} m)	APE (m)	RPE (10^{-2} deg)		RPE (10^{-2} m)	APE (m)	RPE (10^{-2} deg)		RPE (10^{-2} m)	APE (m)	RPE (10^{-2} deg)	
Sequence 26	10.43	**12.75**	**0.8863**		**10.41**	15.094	0.913		11.88	16.546	26.07		11.97	17.743	32.97	
Sequence 29	**11.59**	**19.196**	0.7017		11.77	31.799	**0.6493**		18.1	104.281	25.37		17.12[a]	147.311[a]	19.51[a]	
Sequence 30	**14.42**	**39.16**	**0.38**		38.18	44.271	0.5675		104.8	275.449	38.27		110.5	66.776	16.57	
Sequence 33	**11.27**	**18.187**	**0.6317**		11.38	31.539	0.6717		64.18	209.299	54.68		63.89	71.6447	32.85	
Sequence 35	**42.03**	**5.986**	**0.2057**		61.21	20.493	0.228		187.1	1821.17	15.66		–	–	–	

Best results are in bold

"–" indicates that the algorithm failed during the process

[a]Algorithms have a certain probability of failure in the process of repeated experiments

dynamic objects that are originally static in this sequence, resulting in a slightly lower accuracy than ORBSLAM2.

Sequence 30 is recorded in Gangnam District, which is an important business district in South Korea, the roads in this area are wide and the traffic flow is huge. Therefore, both of the two visual-based algorithms in this sequence have meter-level errors, while IQ-VIO successfully reduces the error to decimeter level.

Sequence 33 is recorded in Yeouido, which is also an important commercial area with large traffic flow. Neither of the two visual-based algorithms, ORBSLAM2 and DynaSLAM, can achieve good results in this sequence.

Sequence 35 is recorded on the highway outside the Seoul, so the algorithm must be initialized in a highly dynamic environment. For DynaSLAM, because it filters the vehicles on the road, it can only track the repeated texture such as lane and guardrail on the road, which soon leads to tracking failure. For ORBSLAM2, there will be a large number of vehicles that keep relatively stationary with themselves on the highway. Therefore, although the algorithm keeps tracking throughout the whole process, huge error occurs. These errors also lead to the decline of the accuracy of traditional EKF algorithm to a certain extent.

IQ-VIO not only ensures the localization accuracy but also the tracking robustness. In most cases, both of ORBSLAM2 and DynaSLAM fail to achieve the ideal accuracy on KAIST dataset. The reason is that, first of all, the frame rate of the image sequence of this dataset is 10 fps, which is slightly lower than the 15 fps required by ORBSLAM2. In addition, the dataset is of dynamic scenes, which further affects the accuracy of the visual algorithm. Finally, the correction matrix of stereo vision provided in this dataset has some deviation, resulting in a little distortion of the image during the final visual tracking, affecting the results of feature extraction and matching, which further affects the accuracy of visual pose estimation. Even so, IQ-VIO still outperforms the other three algorithms in most cases, especially in the high dynamic sequences.

5.4 Indoor experiment

In this experiment, there are moving pedestrians in a corridor with just a few visual features, which is not easy for vision-based localization methods. The purpose of this experiment is to verify the robustness of IQ-VIO compared to other methods under such difficult scenario. The experiment scene is shown in Fig. 15 which is a map built with 3D lidar.

The effect of DynaSLAM at point B is shown in Fig. 16a. Due to the filtering of people in the corridor, the algorithm fails to track at point B. The visual measurement part of IQ-VIO is based on ORBSLAM2, and its tracking effect is shown in Fig. 16b. For ORBSLAM2, it extracts and tracks the features of pedestrians. Although pedestrians, as dynamic

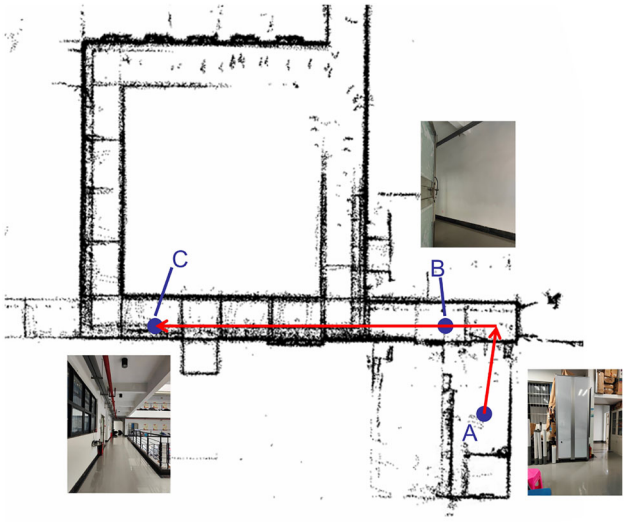


Fig. 15 The corridor scene of indoor experiment. Point A is the start point of the experiment, point B is a place with a few features in the corridor, and point C is the end point of the experiment. The approximate motion trajectory of the robot is shown by the red arrow (colour figure online)

objects in the scene, have a certain impact on the accuracy of visual tracking, they also play the role of visual landmarks in this environment with just a few textures, so the tracking of visual pose can continue. IQ-VIO will not lose tracking in this environment, so the visual odometer section can always provide measurement updates for the filtering algorithm in the subsequent process. Compared with DynaSLAM, IQ-VIO is more robust. Furthermore, due to adaptive fusion, dynamic objects will not affect the accuracy of the final fusion result.

The trajectories generated by each algorithm in the scene of aforementioned indoor corridor are shown in Fig. 17. The black rectangular box is an enlarged view of point B of the corridor, which is the most difficult part with the most sparse texture in the scene. In DynaSLAM, the people in the corridor are filtered, and the scene at this time is a white wall with sparse texture. There are no features that can be used for tracking, which eventually leads to the failure of tracking. In ORBSLAM2, because the pedestrians in the corridor are the only objects with rich texture, the features extraction of ORBSLAM2 focuses on this dynamic objects. That is why ORBSLAM2 drifts seriously here, although it continues to track. In traditional EKF algorithm, the influence of dynamic objects on pose estimation is reduced after filtering, but cannot be completely eliminated. In our algorithm, because the coverage of dynamic objects is greater than the threshold, the confidence of EKF is reduced in the updating process of visual pose measurement, and therefore, a more accurate pose estimation value is finally obtained.

The relative errors of each algorithm on displacement are shown in Fig. 18. DynaSLAM loses visual tracking at about 30 s, and the error of ORBSLAM2 algorithm rises sharply around 30–45 s. Although the traditional EKF algorithm can reduce the error to some extent, the error cannot be completely eliminated. In summary, IQ-VIO completely eliminates the influence of dynamic objects and achieves the best performance. The specific RMSE values of the four algorithms are shown in Table 4.

5.5 Outdoor experiment

The outdoor experiment scene is selected in a university campus. The reference trajectory (red line) is shown in Fig. 19a, and the features matching of DynaSLAM, ORBSLAM2 and IQ-VIO are shown in Fig. 19b–d. This section is the main road leading to the canteen of campus. We deliberately carry out the experiment at nightfall when a large number of students flock to the canteen, which means that the mobile robot is in a real large-scale dynamic scene.

The trajectory generated by each algorithm in this experiment is shown in Fig. 20. Our algorithm achieves the best performance among all other methods. While the dynamic objects in the scene have a significant impact on the pose estimation of ORBSLAM2, so its deviation from the actual trajectory is the most serious.The traditional EKF algorithm and DynaSLAM algorithm are both robust to dynamic objects and their performance is very close.

The displacement RPE of the four algorithms in outdoor experiment is shown in Fig. 21. Obviously, ORBSLAM2 has large deviations in many places. The traditional EKF algo-

Fig. 16 Performances at point B in the corridor



(a) DynaSLAM **(b)** IQ-VIO

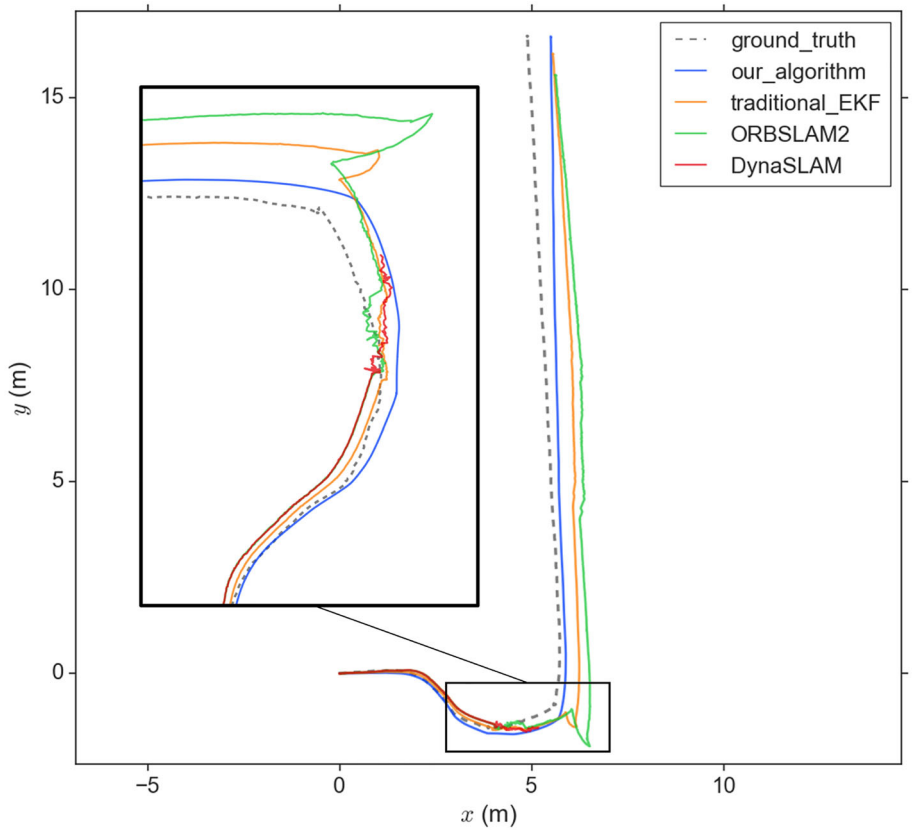


Fig. 17 Trajectories in the indoor experiment

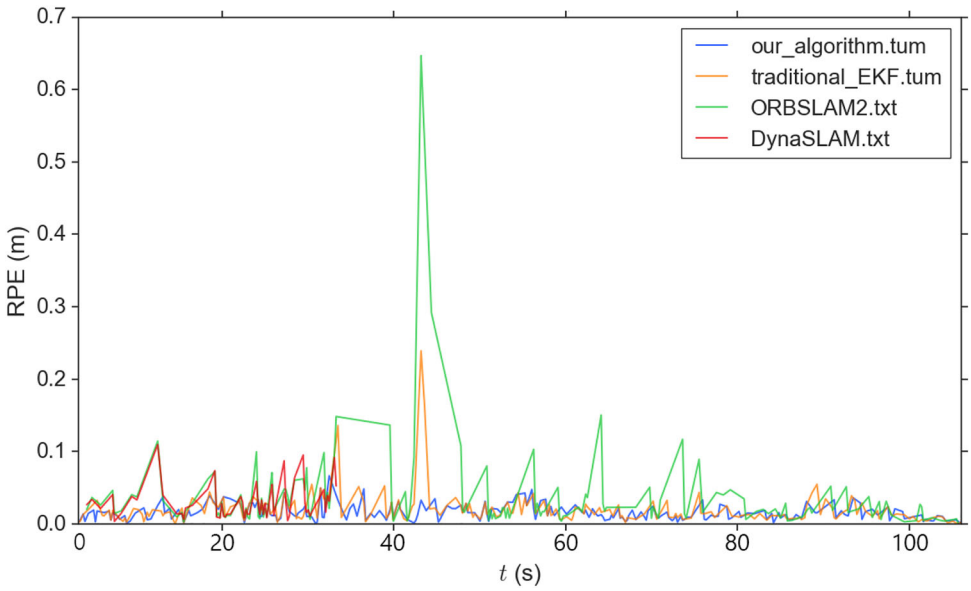


Fig. 18 Displacement RPE in the indoor scene

Table 4 Accuracy error in indoor experiment

Evaluate method (RMSE)	IQ-VIO (ours)	Traditional EKF	ORBSLAM2	DynaSLAM
Displacement RPE (m)	**0.0189**	0.0299	0.0708	–
Angle RPE (°)	**0.378**	0.435	0.647	–
Displacement APE (m)	**0.899**	1.094	1.233	–

Best results are in bold
"–" indicates that the algorithm failed during the process

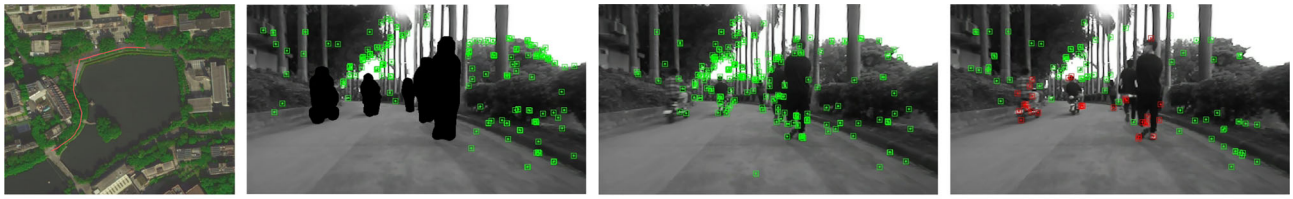


Fig. 19 Outdoor experiment scene. **a** the reference trajectory of outdoor experiment, **b** DynaSLAM, **c** ORBSLAM2, **d** IQ-VIO

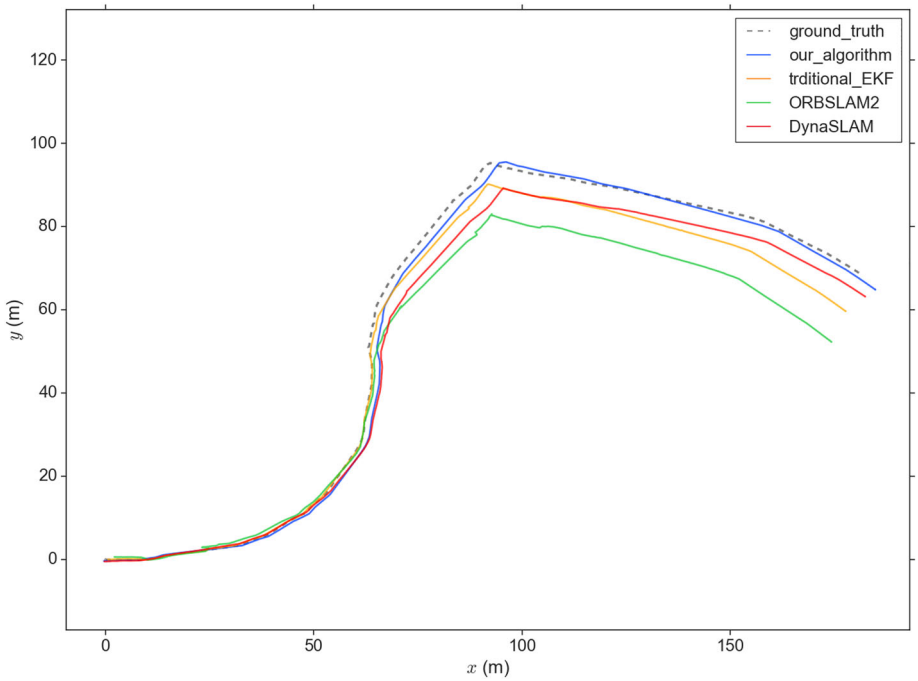


Fig. 20 Trajectories in the outdoor experiment

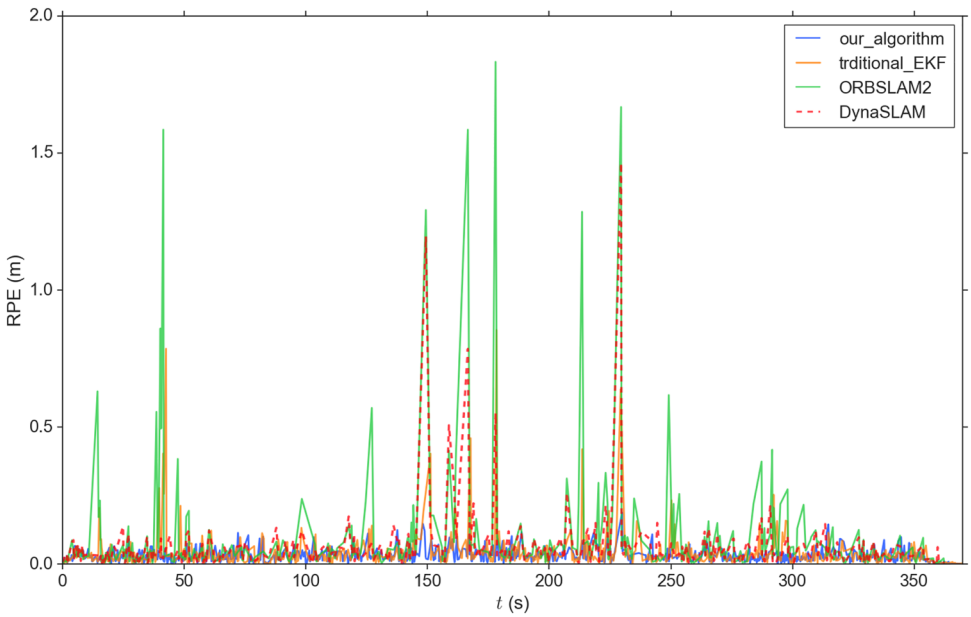


Fig. 21 Displacement RPE in the outdoor scene

Table 5 Accuracy error in outdoor experiment

Evaluate method (RMSE)	IQ-VIO (ours)	Traditional EKF	ORBSLAM2	DynaSLAM
Displacement RPE (m)	**0.0381**	0.0765	0.225	0.125
Angle RPE (°)	**0.212**	0.218	0.361	0.323
Displacement APE (m)	**3.032**	4.845	9.521	4.241

Best results are in bold

rithm can reduce these deviations to some extent, but cannot completely remove them. DynaSLAM has large deviations in a few cases. Our algorithm maintains a low error level throughout. The specific RMSE values of the four algorithms are shown in Table 5. There is no significant difference in the angle RPE of the four algorithms.

6 Conclusion

We propose an adaptive visual inertial odometry localization method under dynamic environments. Our algorithm recognizes the dynamic objects in the scene through the target detection network, and then extracts and matches the features of the image through the visual tracking algorithm. We introduce the feature coverage which represents the proportion of the area covered by the dynamic objects in the image. Furthermore, a dynamic scene interference index is defined by the coverage of features on the dynamic object, so as to dynamically change the measurement error covariance matrix of the EKF, and finally eliminate the impact of the dynamic objects on the fusion positioning results. Experiments show that our adaptive algorithm can get reliable localization in high dynamic environments. Compared with the other three algorithms, it has achieved the best results in both accuracy and robustness in dynamic scenes.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by HZ, KL and JX. The first draft of the manuscript was written by HZ and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The authors make sure that all data and materials as well as software application or custom code support their published claims and comply with field standards.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical statement and informed consent Informed consent was obtained from all individual participants included in the study.

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